Assignment 2

EE5141 - Wireless Communication

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1 Introduction

This report contains the answers to assignment 2. It only contains my plots and approaches to the problems.The code to generate these plots are in the python file which I have submitted.

2 Question 1

2.1 Part a - Schmidl Cox algorithm

The preamble required for the schimdl cox algorithm needs to by symmetric in the time domain. If we look at the N length signal(without the CP) x[n] we have.

$$x[n] = x\left[n - \frac{N}{2}\right], \quad \forall n \in \left\{\frac{N}{2}, \frac{N}{2}+1, \ldots, N-1\right\}$$

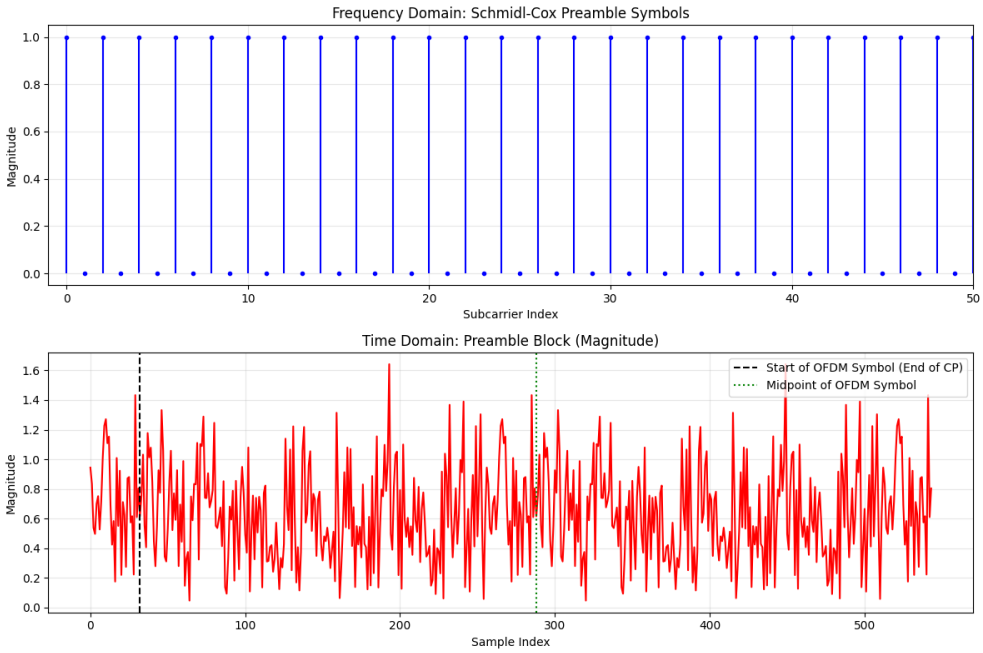


Figure 1: Schmidl Cox preamble in time and frequency

2.1.1 Derivation of Frequency Domain

The corresponding frequency domain for this will be present in alternate subcarriers.This comes from.

$$\text{Given the symmetry condition: } x[n] = x\left[n + \frac{N}{2}\right], \quad \forall n \in \left\{0, \ldots, \frac{N}{2} - 1\right\}$$

$$\text{The DFT is defined as: } X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

$$\text{Splitting the summation: } X[k] = \sum_{n=0}^{\frac{N}{2}-1} x[n] e^{-j\frac{2\pi}{N}kn} + \sum_{n=\frac{N}{2}}^{N-1} x[n] e^{-j\frac{2\pi}{N}kn}$$

$$\text{Substituting } m = n - \frac{N}{2} \text{ in the second sum: } X[k] = \sum_{n=0}^{\frac{N}{2}-1} x[n] e^{-j\frac{2\pi}{N}kn} + \sum_{m=0}^{\frac{N}{2}-1} x\left[m + \frac{N}{2}\right] e^{-j\frac{2\pi}{N}k\left(m+\frac{N}{2}\right)}$$

$$\text{Using the symmetry property } x[m] = x\left[m + \frac{N}{2}\right] : X[k] = \sum_{n=0}^{\frac{N}{2}-1} x[n] e^{-j\frac{2\pi}{N}kn} \left[1 + e^{-j\frac{2\pi}{N}k\frac{N}{2}}\right]$$

$$\text{Simplifying the exponential: } X[k] = \sum_{n=0}^{\frac{N}{2}-1} x[n] e^{-j\frac{2\pi}{N}kn} \left[1 + e^{-j\pi k}\right]$$

$$\text{For odd } k, \quad e^{-j\pi k} = -1 \implies X[k] = 0$$

2.1.2 Derivation of max frequency offset correction

Let $y[n] = x[n]e^{j2\pi\Delta f n T_s}$ be the received sequence in the presence of a carrier frequency offset Δf. The Schmidl-Cox correlation metric $R[n]$ is calculated as:

$$\begin{aligned}
R[n] &= \sum_{i=-N/2+1}^{0} y[n-i]y^*[n-i-N/2] \\
&= \sum_{i=-N/2+1}^{0} \left(x[n-i]e^{j2\pi\Delta f(n-i)T_s}\right)\left(x^*[n-i-N/2]e^{-j2\pi\Delta f(n-i-N/2)T_s}\right) \\
&= \sum_{i=-N/2+1}^{0} x[n-i]x^*[n-i-N/2] \cdot e^{j2\pi\Delta f T_s[(n-i)-(n-i-N/2)]} \\
&= \left(\sum_{i=-N/2+1}^{0} |x[n-i]|^2\right) e^{j\pi\Delta f N T_s}
\end{aligned}$$

The phase of the correlation metric is derived as:

$$\angle R[n] = \pi\Delta f N T_s \pmod{2\pi}$$

To resolve the frequency offset without phase ambiguity, the phase must be restricted to the principal interval of the arctangent function $(-\pi, \pi]$:

$$-\pi < \pi \Delta f N T_s \leq \pi$$
$$-\frac{1}{NT_s} < \Delta f \leq \frac{1}{NT_s}$$

Defining the subcarrier spacing as $\Delta f_{sc} = \frac{1}{NT_s}$, the maximum frequency offset that can be correctly estimated is:

$$|\Delta f| \leq \Delta f_{sc}$$

Therefore for the give problem only frequency offset of -10Khz to 10Khz can be estimated.

2.2 part b - Frequency Offset estimate using schmidl cox

Since we can only estimate frequency offset between -10 to 10kHz I have taken the frequency offset to be 5kHz and plotted MSE vs SNR.

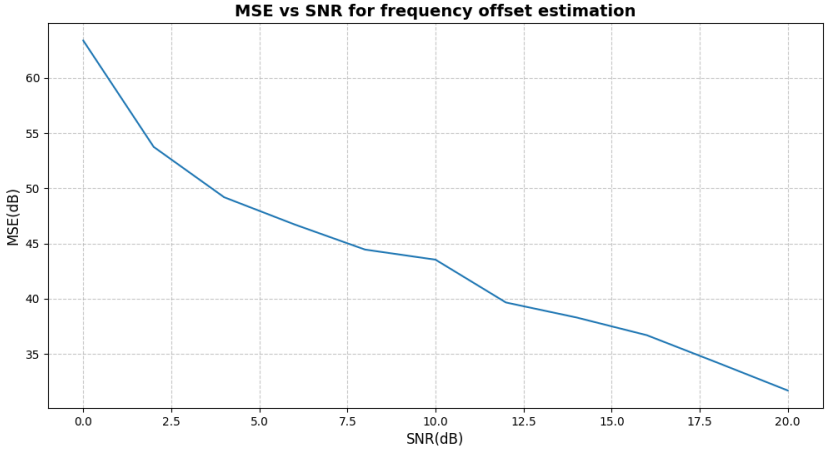


Figure 2: MSE vs SNR for frequency offset estimation

3 Question 2

Given below is the plot for schmidl cox correlation for the different channels at h=1 and given h in Q1 for SNR of 6dB.I have found the first point after which the peak reduces to estimate the start of the frame.The noise is way too high.

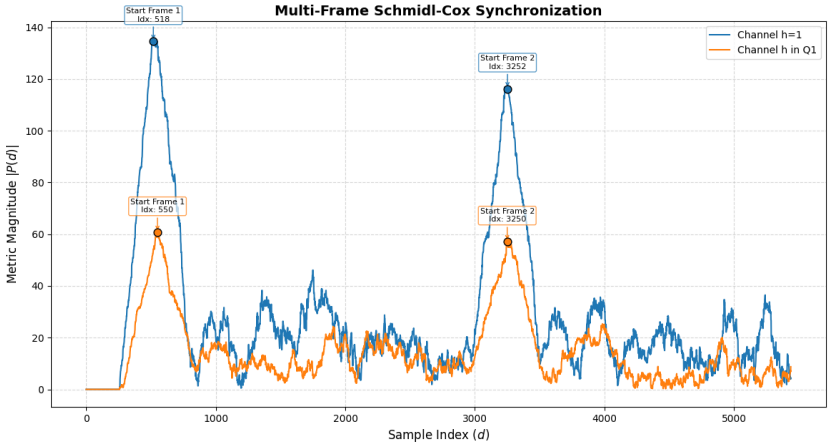


Figure 3: Schmidl Cox Synchronisation

4 Question 3

Find below my graph of MSE vs SNR for various algorithms for channel estimation.
NOTE: The ZF estimate is only done on the pilot locations.
For the windowing based FFT method I have done a hanning window for the last 10 taps for my Ncp size channel filter.
For the e part if we know the channel locations then we can pick those columns of the DFT matrix and solve the least squares solution. Once we get the least squares solution we then pad zeroes in all location other than the locations we know and take an fft to get the channel frequency response. You can see the plot below that it is much better than just doing ordinary least squares on Ncp locations.

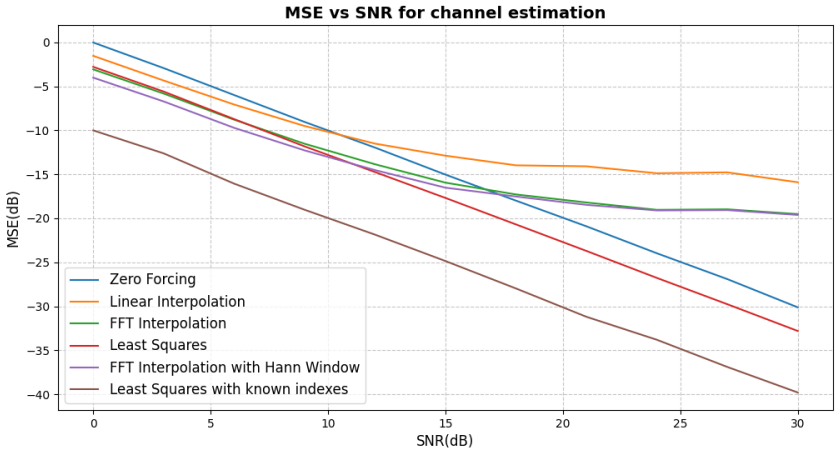


Figure 4: MSE vs SNR for different channel estimates